Ojasva Nema

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Education

Indian Institute of Technology Roorkee *2023 – 2027*
B.Tech, Metallurgical and Materials Engineering | CGPA: 8.71 *Roorkee, India*

Publications & Preprints

Mechanistic Evidence for Spectral Structures in Prior-Data Fitted Networks *2026*
***Ojasva Nema**, Kaustubh Sharma, Srijan Tiwari, Parikshit Pareek* *ICML Mech. Interp. workshop 2026*
(PFN attention latents encode a spectral manifold; we can decode GP kernels in one forward pass.)

Structural Disentanglement in Bilinear MLPs via Architectural Inductive Bias *2026*
***Ojasva Nema**, Kaustubh Sharma, Aditya Chauhan, Parikshit Pareek* *Under review at ICLR 2027*
(Bilinear MLPs enable surgical unlearning without disrupting retained tasks.)

Localised Routing, Distributed Execution: How LLMs Process Semantic Domains *2026*
***Ojasva Nema**, Kaustubh Sharma, Abhiraj Bharangar, Manjot Singh, Srijan Tiwari* *Under review at ICLR 2027*
(Proved via causal patching that localized attention routes semantic domains for distributed MLP execution.)

Sparsity-Aware Representation Learning for Jet Image Generation via Guided Latent Diffusion *2025*
***Ojasva Nema**, Kaustubh Sharma, Parikshit Pareek* *Code | Preprint*
(Modeled a physics-aware diffusion model for jet image synthesis at CERN's Large Hadron Collider.)

Image-Alchemy: Personalized Text-to-Image Generation *2025*
***Ojasva Nema**, Kaustubh Sharma, Cherish Puniani, Amritanshu Tiwari* *ICLR DeLTa Workshop*
(Developed a pipeline to personalize Stable Diffusion XL with high quality subject integration.)

Research & Professional Experience

Summer Research Intern at Yale University *May 2026 – Jul 2026*
Causal Evidence & Decisions Studio, Prof. Harsh Parikh *New Haven, USA*

- Formulating a theoretical framework to construct Confidence Sets of Representations.
- Implemented and evaluated multiple causal representation learning algorithms, applying structural relaxations to bound the unidentifiable representation space.
- Overcomes the limitations of purely predictive models by recovering true causal mechanisms, enabling robust counterfactual reasoning and scientific discovery.

Undergraduate Research Assistant at P-Square Lab *Sep 2025 – Present*
P-Square Lab, Prof. Parikshit Pareek, IIT Roorkee *Roorkee, India*

- Demonstrated orthogonal task structures induce disentanglement in MLPs, enabling surgical unlearning.
- Designed amortized inference for zero-shot Gaussian Process kernel discovery from limited data using PFNs.
- Engineered a sparsity-aware latent diffusion model for ultra-sparse jet image synthesis in HEP.

Interpretable Bayesian Regression for Tabular Data *Oct 2025 – Jan 2026*
Data Science Group, IIT Roorkee *Roorkee, India*

- Built a Decoder to extract Gaussian Process kernels in 3.6 ms, achieving a 250x speedup over current baselines.
- Executed downstream Bayesian optimization entirely on CPU with 4.8 ms latency, matching ground-truth accuracy.
- Demonstrated core tabular features form in early layers, enabling 12x model compression with 3.2x faster inference.

AI Engineer (Intern) at Elimentary (YC F26) *Nov 2025 – Jan 2026*
Elimentary *Bangalore, India*

- Built secure, quantized local LLM APIs, achieving granular layer-wise observability.
- Optimized GPU throughput by implementing sequence scheduling for concurrent LLM inference requests.
- Designed a no-code LLM evaluation platform integrated with Langfuse and Grafana for comprehensive monitoring.

Achievements

- **Google DeepMind APAC Research Symposium 2026:** Selected as a student research delegate.
- **EY Techathon 6.0:** Secured a **Top-3** finish among **180,000+** participants nationwide.
- **Amazon ML Challenge 2025:** Ranked **66th** out of **82,789** participants in a national ML competition.
- **STC-UP Hackathon:** Ranked **Top 11** across India; only undergraduate team competing against industry startups.